2022-2 Calculus II (06)

Quiz 2

2022.11.22, 15:05–15:45

5 problems. All your solutions should contain rigorous explanation.

1. (10pts) Evaluate the triple integral
$$\iiint_E z\,dV,$$
where E is bounded by the cylinder $y^2+z^2=9$ and the planes $x=0$, $y=3x$, and $z=0$ in the first octant.

2. (10pts) Evaluate the integral
$$\int_0^1\int_0^{\sqrt{1-x^2}}\int_{\sqrt{x^2+y^2}}^{\sqrt{2-x^2-y^2}} xy\,dz\,dy\,dx$$
by changing to spherical coordinates.

3. (10pts) Evaluate the triple integral
$$\iiint_E xy\,dV,$$
where E is the region that lies between the cone $z=\sqrt{x^2+y^2}$ and the paraboloid $z=6-x^2-y^2$.

4. (10pts) Evaluate the double integral
$$\iint_R (4x+8y)\,dA,$$
where R is the paralellogram with vertices $(-1,3)$, $(1,-3)$, $(3,-1)$, and $(1,5)$.

5. (10pts) Evaluate the line integral
$$\int_C (y+z)\,dx+(x+z)\,dy+(x+y)\,dz,$$
where C consists of line segments from $(0,0,0)$ to $(1,0,1)$ and from $(1,0,1)$ to $(0,1,2)$.